

AeroDex: Mathematical & Econometric Manual

National Airfare Price Index (APIx) • Smart India Hackathon 2026 (Problem Statement: SIH26056)

1. National Composite Laspeyres Price Index

The core computational engine of AeroDex implements the fixed-basket Laspeyres aggregation standard prescribed by the National Statistical Office (NSO) and the International Monetary Fund (IMF) Consumer Price Index Manual (2020).

EQUATION 1.1: NATIONAL COMPOSITE LASPEYRES INDEX

$$I_{APIx,t} = \frac{\sum_{r=1}^R P_{r,t} \cdot Q_{r,0}}{\sum_{r=1}^R P_{r,0} \cdot Q_{r,0}} \times 100 = \sum_{r=1}^R W_r \cdot \left(\frac{P_{r,t}}{P_{r,0}} \right) \times 100$$

- R : Total scheduled domestic city-pair corridors ($R = 786$ nationwide; 25 top core corridors sampled live).
- $P_{r,t}$: Carrier-market-share-weighted representative economy tariff on corridor r observed at time t .
- $P_{r,0}$: Baseline calibrated economy tariff on corridor r for the base period (2024 = 100.0).
- $Q_{r,0}$: Base year annual passenger census volume on corridor r (totaling 136.0 Million flyers).
- W_r : Spatial expenditure weight of corridor r relative to total national passenger air expenditure.

EQUATION 1.2: SINGLE-ROUTE PRICE RELATIVES

$$I_{r,t} = \left(\frac{P_{r,t}}{P_{r,0}} \right) \times 100$$

- Represents the price index of an individual route relative to base year 2024.

2. Spatial Passenger Census Weighting (DGCA Form-A)

Weights are derived strictly from official Directorate General of Civil Aviation (DGCA) Form-A annual origin-destination traffic statistics, guaranteeing democratic spatial representation across national trunk, metro, and regional connectivity corridors.

EQUATION 2.1: ROUTE EXPENDITURE WEIGHT

$$W_r = \frac{Q_{r,0} \cdot P_{r,0}}{\sum_{k=1}^R Q_{k,0} \cdot P_{k,0}}, \quad \text{subject to} \quad \sum_{r=1}^R W_r = 1.0 \quad (100\%)$$

- Ensures heavy corridors (e.g. DEL-BOM, BLR-DEL) influence headline national inflation proportionately to flyer expenditure.

Corridor Code	City Pair	Annual Passengers	Corridor Weight (W_r)	Cumulative Weight
DEL-BOM	Delhi ↔ Mumbai	11,128,450	8.18%	8.18%
BLR-DEL	Bengaluru ↔ Delhi	7,389,120	5.43%	13.61%
DEL-BLR	Delhi ↔ Bengaluru	6,274,310	4.61%	18.22%
BOM-BLR	Mumbai ↔ Bengaluru	5,892,100	4.33%	22.55%
DEL-CCU	Delhi ↔ Kolkata	4,912,400	3.61%	26.16%
DEL-HYD	Delhi ↔ Hyderabad	4,680,200	3.44%	29.60%

3. Carrier Market Share Capacity Aggregation

Corridor-level tariffs are not naive arithmetic means; they are weighted by each scheduled airline's capacity share from official DGCA monthly census returns.

EQUATION 3.1: CARRIER-WEIGHTED REPRESENTATIVE FARE

$$P_{r,t} = \frac{\sum_{c \in C_r} \omega_c \cdot \bar{p}_{r,c,t}}{\sum_{c \in C_r} \omega_c}$$

- C_r : Set of commercial scheduled airlines operating non-stop services on route r .
- $\bar{p}_{r,c,t}$: Median economy tariff quoted by airline c on route r at observation time t .
- ω_c : Airline domestic capacity market share (IndiGo: 61.2%, Air India: 14.3%, AI Express: 6.4%, Akasa: 4.8%, SpiceJet: 4.2%, Others: 9.1%).

4. Statutory Tariff Deconstruction Protocol

In accordance with DGCA Tariff Monitoring Unit (TMU) directives and Ministry of Civil Aviation regulations, raw gross tariffs are unbundled to isolate carrier dynamic pricing from fuel surcharges and statutory airport fees.

EQUATION 4.1: STATUTORY TARIFF DECOMPOSITION

$$P_{\text{Total}} = P_{\text{Base}} + P_{\text{YQ}} + P_{\text{Airport}} + P_{\text{GST}}$$

$$P_{\text{Base}} = \frac{P_{\text{Total}} - P_{\text{Airport}}}{1.05} \times 0.795$$

$$P_{\text{YQ}} = \frac{P_{\text{Total}} - P_{\text{Airport}}}{1.05} \times 0.205$$

$$P_{\text{GST}} = 0.05 \times (P_{\text{Base}} + P_{\text{YQ}})$$

- P_{Base} : Carrier net inventory revenue ($\approx 70\%$ of gross fare).
- P_{YQ} : Fuel Surcharge (pass-through of IOCL Jet Fuel ATF prices, $\approx 18\%$ of gross fare).
- P_{Airport} : User Development Fee (UDF) and Passenger Service Fee (PSF) ($\approx 7\%$ of gross fare).
- P_{GST} : Statutory Ministry of Finance Goods and Services Tax (5% on domestic economy).

Numerical Validation Example:

For gross coach fare $P_{\text{Total}} = ₹7,200$ with standard airport fee $P_{\text{Airport}} = ₹504$:

Net taxable component = $(7,200 - 504)/1.05 = ₹6,377.14$

$P_{\text{Base}} = 6,377.14 \times 0.795 = ₹5,069.83$ • $P_{\text{YQ}} = 6,377.14 \times 0.205 = ₹1,307.31$ • $P_{\text{GST}} = (5,069.83 + 1,307.31) \times 0.05 = ₹318.86$

Reconciliation: $5,069.83 + 1,307.31 + 504.00 + 318.86 = ₹7,200.00$ (Residual: ₹0.00).

5. Macroeconomic Linkage: Headline CPI Sensitivity

Enables the Reserve Bank of India (RBI) Monetary Policy Committee (MPC) and MoSPI to instantly measure the basis-point impact of domestic airfare inflation on the Combined All-India Consumer Price Index.

EQUATION 5.1: HEADLINE CPI IMPACT IN BASIS POINTS

$$\Delta \text{CPI}_{\text{bps}} = \left(\frac{I_{\text{APIx},t} - 100.0}{100.0} \right) \times w_{\text{Airfare}} \times 10,000$$

- $w_{\text{Airfare}} = 0.00077$: Official weight (0.077%) of passenger air transport in the MoSPI CPI basket (COICOP 07.3.3).
- 10,000: Multiplier converting fractional inflation changes into basis points ($1\% = 100 \text{ bps}$).

RBI MPC Application:

If the AeroDex index registers 133.78 (+33.78% increase over the 2024 baseline):

$$\Delta \text{CPI}_{\text{bps}} = 0.3378 \times 0.00077 \times 10,000 = +2.60 \text{ basis points} \quad (+0.026\%)$$

6. Statistical Data Cleaning: 2.5x Median & Tukey IQR

To prevent business class listings and cancellation artifacts from distorting the economy baseline, AeroDex applies a robust non-parametric trimming protocol.

EQUATION 6.1: NON-PARAMETRIC 2.5X MEDIAN RULE

$$\text{Threshold}_{\text{Upper}} = \max(\text{₹}28,000, 2.5 \times \text{Median}(P_r))$$
$$\text{Valid Quote Set} = \{p_i \mid \text{₹}1,500 \leq p_i \leq \text{Threshold}_{\text{Upper}}\}$$

EQUATION 6.2: TUKEY 1.5X INTERQUARTILE RANGE (IQR) CLEANLINESS BOUNDS

$$\text{IQR} = Q_3 - Q_1$$
$$\text{Lower Fence} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper Fence} = Q_3 + 1.5 \times \text{IQR}$$

- Q_1, Q_3 : First and third quartiles of corridor fare distribution. Eliminates spurious algorithmic spikes.

7. Real-Time Econometric Data Integrity Score

Quantifies microdata reliability (0 – 100, Grade A+) before quotes enter official econometric indices.

EQUATION 7.1: 4D COMPOSITE INTEGRITY SCORE

$$\text{Integrity Score} = 0.35 \cdot S_{\text{fidelity}} + 0.25 \cdot S_{\text{outlier}} + 0.20 \cdot S_{\text{diversity}} + 0.20 \cdot S_{\text{freshness}}$$

- **Fare Split Fidelity** (S_{fidelity}): Precision of reconciliation $|P_{\text{Total}} - (P_{\text{Base}} + P_{\text{YQ}} + P_{\text{Airport}} + P_{\text{GST}})| \leq ₹1.00$.
- **Statistical Cleanliness** (S_{outlier}): Percentage of quotes conforming to Tukey $1.5 \times \text{IQR}$ bounds.
- **Source Diversity** ($S_{\text{diversity}}$): Ratio of unique OTAs and direct airlines contributing live quotes.
- **Temporal Freshness** ($S_{\text{freshness}}$): $S_{\text{freshness}} = 100 \times e^{-\lambda \cdot \Delta t}$, where $\lambda = \frac{\ln(2)}{180 \text{ min}}$ (3-hour half-life).

8. Superlative Fisher Ideal Index & Substitution Bias

Used in the Historical Crisis Simulation Studio (May 2023 Go First Grounding, April 2019 Jet Airways Collapse) to demonstrate how fixed-basket Laspeyres overstates inflation during major supply shocks.

EQUATION 8.1: FISHER IDEAL INDEX & SUBSTITUTION BIAS

$$I_{\text{Laspeyres}} = \frac{\sum P_t \cdot Q_0}{\sum P_0 \cdot Q_0} \times 100, \quad I_{\text{Paasche}} = \frac{\sum P_t \cdot Q_t}{\sum P_0 \cdot Q_t} \times 100$$

$$I_{\text{Fisher}} = \sqrt{I_{\text{Laspeyres}} \times I_{\text{Paasche}}}$$

$$\text{Substitution Bias} = I_{\text{Laspeyres}} - I_{\text{Fisher}}$$

- *Empirical Grounding Result:* Go First Grounding caused $I_{\text{Laspeyres}} = 188.4$ vs $I_{\text{Fisher}} = 173.6$, demonstrating +14.8 index points substitution bias.

9. Econometric Nowcasting & Calibrated Multipliers

Generates high-frequency expected tariffs across advance booking windows ($T + 1$ to $T + 45$) with weather and festival shock adjustments.

EQUATION 9.1: MULTI-FACTOR PREDICTIVE TARIFF ENGINE

$$\hat{P}_{r,T} = f_{\text{RF}}(T_{\text{advance}}, \text{Dist}_{\text{km}}, \text{Month}, \text{IsWeekend}) \times M_{\text{event}} \times M_{\text{calamity}}$$

- f_{RF} : Random Forest Regressor trained on 14,500+ historical microdata records.
- $M_{\text{event}} \in [1.20, 1.90]$: Calibrated festival multiplier (Diwali +65%, Chhath +85%, Durga Puja +45%).
- $M_{\text{calamity}} \in [1.50, 1.80]$: Calamity capacity factor triggered by UN GDACS RSS alerts or NOAA METAR fog alerts.

10. Temporal Rate of Change Formulations

EQUATION 10.1: WEEK-OVER-WEEK (WOW) INDEX GROWTH

$$\text{WoW}_{\%} = \left(\frac{I_{APIx,w} - I_{APIx,w-1}}{I_{APIx,w-1}} \right) \times 100$$

EQUATION 10.2: INTRADAY COEFFICIENT OF VARIATION (VOLATILITY CV)

$$\text{CV}_{r,\text{day}} = \left(\frac{\sigma_{P_r}}{\mu_{P_r}} \right) \times 100$$

- Measures intraday dynamic pricing volatility across peak vs off-peak departure banks.